

★ WEEK 5 · PART 2 of 3

GARCH and the Volatility Toolkit

The Workhorse That Runs the World's Risk Desks

GARCH(1,1) · persistence & mean reversion · fat tails · the leverage effect

ARCH captured clustering but needed too many lags. GARCH fixes that with one idea — a memory of past variance — and, with two extensions, matches every stylized fact of returns.

2

PART 2 • GARCH

The Volatility Workhorse

GARCH's variance memory, the persistence that governs mean reversion, volatility forecasting, and the extensions for fat tails and the leverage effect.

One Idea Away From the Workhorse

ARCH pointed the way; GARCH completes it

1

ARCH captured clustering

Tying variance to recent squared shocks reproduced clustering and fat tails — the first real model of time-varying risk.

2

But it was inefficient

Volatility's long memory forced a high ARCH order and many parameters — noisy, hard to estimate, prone to over-fitting.

3

GARCH adds memory of past variance

One extra term — yesterday's variance — captures that long memory compactly. GARCH(1,1) does what ARCH(10+) struggled to.

Part 2 in one line: add a variance-memory term, and three parameters model the world's risk.

The GARCH(p,q) Model

Past shocks AND past variance

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2 \quad \text{GARCH}(p, q)$$

What is new

- The variance now also depends on its own recent past.
- The β terms carry variance forward day after day.
- p counts variance lags; q counts squared-shock lags.

Why it helps

- Persistence is captured by β , not by piling up ARCH lags.
- Far fewer parameters for the same long memory.
- Almost always, $p = q = 1$ is all you need.

Why Add a Memory of Past Variance?

One term replaces many ARCH lags

A GARCH(1,1) is equivalent to an ARCH with infinitely many lags — the single β term compresses all of that long memory into one parameter.

The problem it solves

- Volatility clusters persist for weeks, not days.
- Pure ARCH needs one lag per day of memory.
- That is wasteful and statistically fragile.

The elegant fix

- $\beta \cdot$ (yesterday's variance) already contains all earlier variance.
- So one recursive term reaches arbitrarily far back.
- Compact, stable, and easy to estimate.

GARCH(1,1): The Workhorse

Three parameters that run the world's risk desks

$$\sigma_t^2 = \underbrace{\omega}_{\text{baseline}} + \underbrace{\alpha \varepsilon_{t-1}^2}_{\text{reaction}} + \underbrace{\beta \sigma_{t-1}^2}_{\text{persistence}} \quad \text{GARCH}(1,1)$$

The three ingredients

- ω : a baseline that anchors the long-run variance level.
- α : the reaction to the most recent shock.
- β : the persistence — how much of yesterday's variance carries over.

Why it dominates

- It fits an enormous range of assets with just three numbers.
- It is the industry default for VaR and option pricing.
- Everything fancier is a variation on this core.

The GARCH(1,1) Recipe

Today's variance is a baseline (ω), plus a reaction to the last shock ($\alpha \cdot \text{shock}^2$), plus a memory of yesterday's variance ($\beta \cdot \text{prev}$). Three ingredients, one recursive rule.

The GARCH(1,1) Recipe

Today's variance = a baseline + a reaction to the last shock + a memory of yesterday's variance.

$\sigma^2_t =$

omega

baseline variance
(the long-run anchor)

+

alpha $e^2_{(t-1)}$

reaction to last shock
(short-run, twitchy)

+

beta $s^2_{(t-1)}$

yesterday's variance
(memory, persistence)

Why beta is the breakthrough

- beta carries variance forward day after day
- one term captures LONG memory that ARCH needs many lags to reproduce
- GARCH(1,1) ~ an infinite ARCH, compactly

The three-parameter workhorse

- omega, alpha, beta: just three numbers
- alpha + beta = persistence (near 1 in practice)
- it runs risk desks, VaR, and option models
- the default first choice for any return series

Persistence and Mean Reversion

What $\alpha + \beta$ tells you

$$\text{persistence} = \alpha + \beta, \quad \bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta} \quad (\alpha + \beta < 1)$$

Reading $\alpha + \beta$

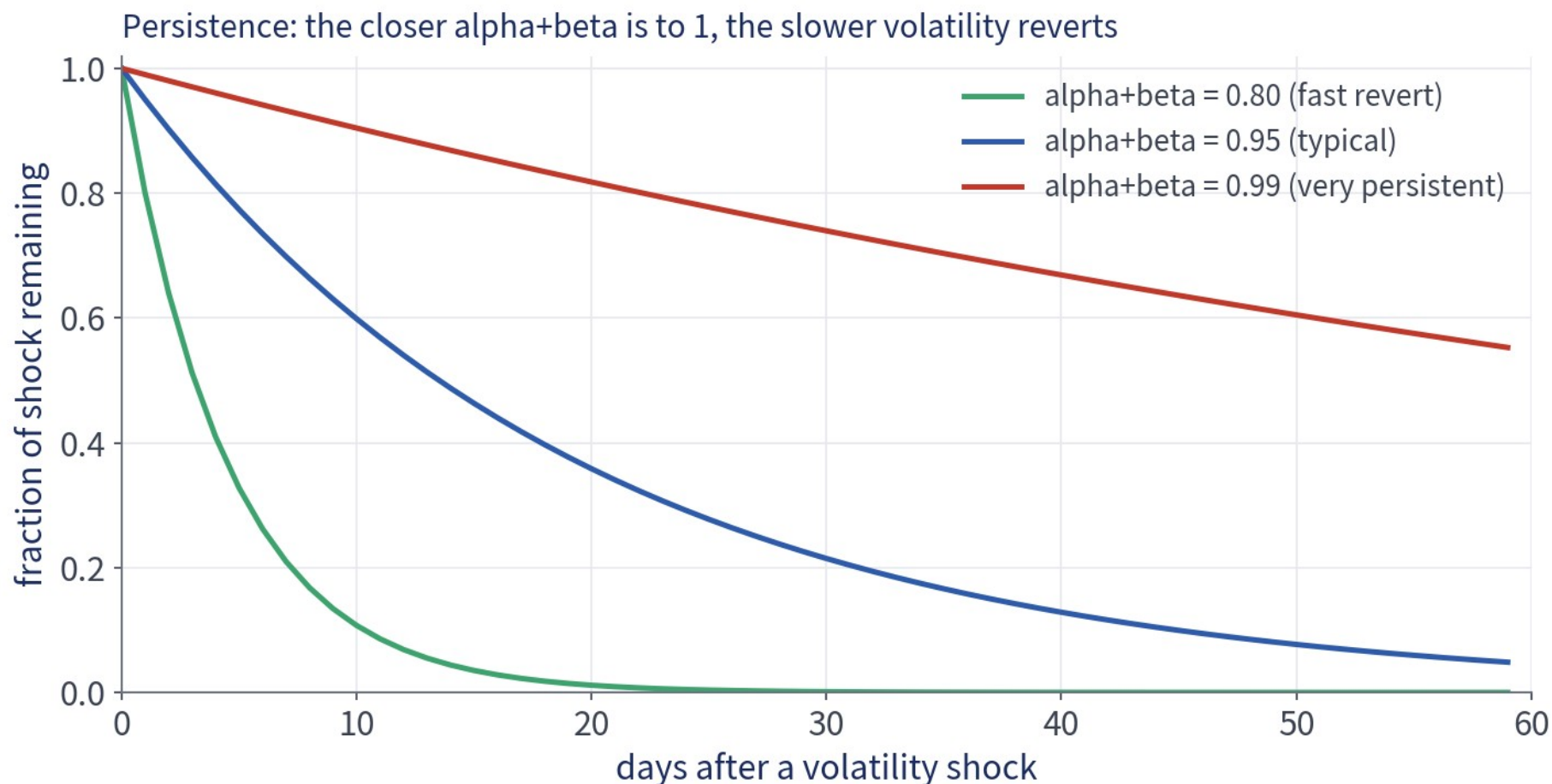
- Their sum is the persistence of a volatility shock.
- Close to 1 (often 0.95–0.99) means very slow reversion.
- The unconditional variance is $\omega/(1 - \alpha - \beta)$.

Why it matters

- It tells you how long a spike in risk will linger.
- It must stay below 1, or variance is not well-defined.
- At exactly 1 (IGARCH), shocks to variance are permanent.

How Fast Does a Volatility Shock Fade?

The closer $\alpha + \beta$ is to 1, the slower a shock to volatility decays. Typical equity estimates near 0.95–0.99 mean risk stays elevated for weeks after a storm.



Volatility Mean-Reverts

Storms and calm both eventually pass

Because $\alpha + \beta$ is below 1, the conditional variance is always pulled back toward its long-run level $\omega/(1 - \alpha - \beta)$. High volatility subsides; unusual calm eventually ends.

What it means

- After a spike, forecast volatility drifts back down over time.
- After deep calm, it drifts back up.
- The long-run level is the anchor it orbits.

Why traders rely on it

- It makes multi-day volatility forecasts sensible.
- It underlies the term structure of implied volatility.
- It is the reason a VIX spike is expected to fade.

The Volatility Half-Life

Turning persistence into a plain-English horizon

How to read it

- Higher persistence → a longer half-life.
- $\alpha + \beta = 0.95$ gives a half-life of roughly two weeks.
- $\alpha + \beta = 0.99$ stretches that to over two months.

Why it is handy

- It communicates risk persistence to non-specialists.
- It sets a sensible horizon for risk limits and hedges.
- It sanity-checks an estimate: is the half-life plausible?

Forecasting Volatility

Tomorrow's risk, and beyond

$$\mathbb{E}_t[\sigma_{t+h}^2] = \bar{\sigma}^2 + (\alpha + \beta)^{h-1} (\sigma_{t+1}^2 - \bar{\sigma}^2) \xrightarrow{h \rightarrow \infty} \bar{\sigma}^2$$

One and many steps

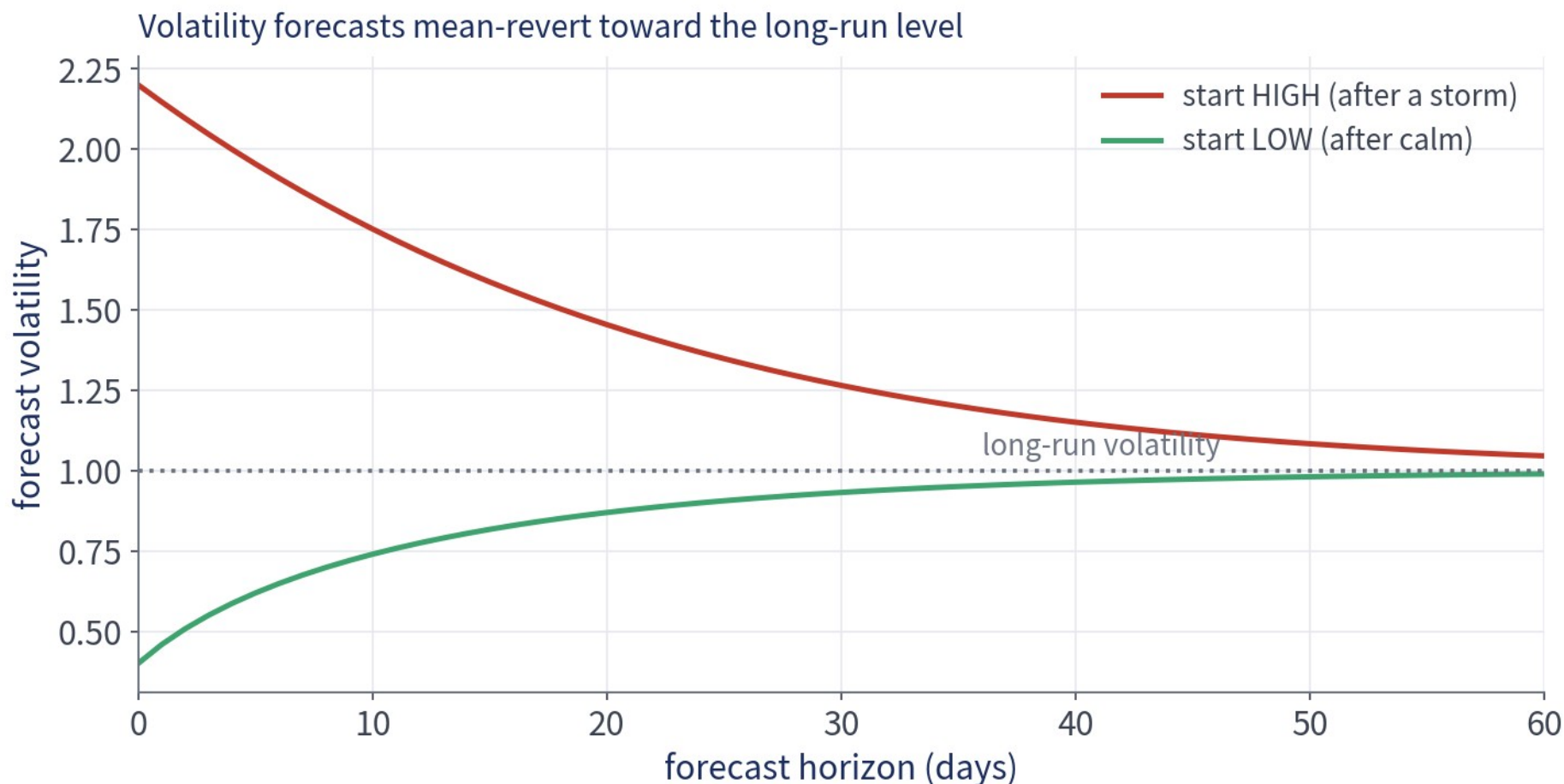
- One step ahead follows directly from the GARCH equation.
- Multi-step forecasts decay geometrically toward the long-run level.
- The speed of decay is governed by $\alpha + \beta$.

The term structure

- Near horizons reflect today's conditions.
- Far horizons converge to the unconditional variance.
- This gives volatility its own 'term structure'.

The Volatility Forecast Term Structure

From a high starting point (after a storm) the forecast decays toward the long-run level; from a low one it rises toward it. Both converge to the same anchor.



Fat Tails: Student-t Innovations

GARCH with normal shocks is not heavy-tailed enough

$$\varepsilon_t = \sigma_t z_t, \quad z_t \sim t_\nu \text{ (standardized) } \quad \text{heavier tails than normal}$$

The problem

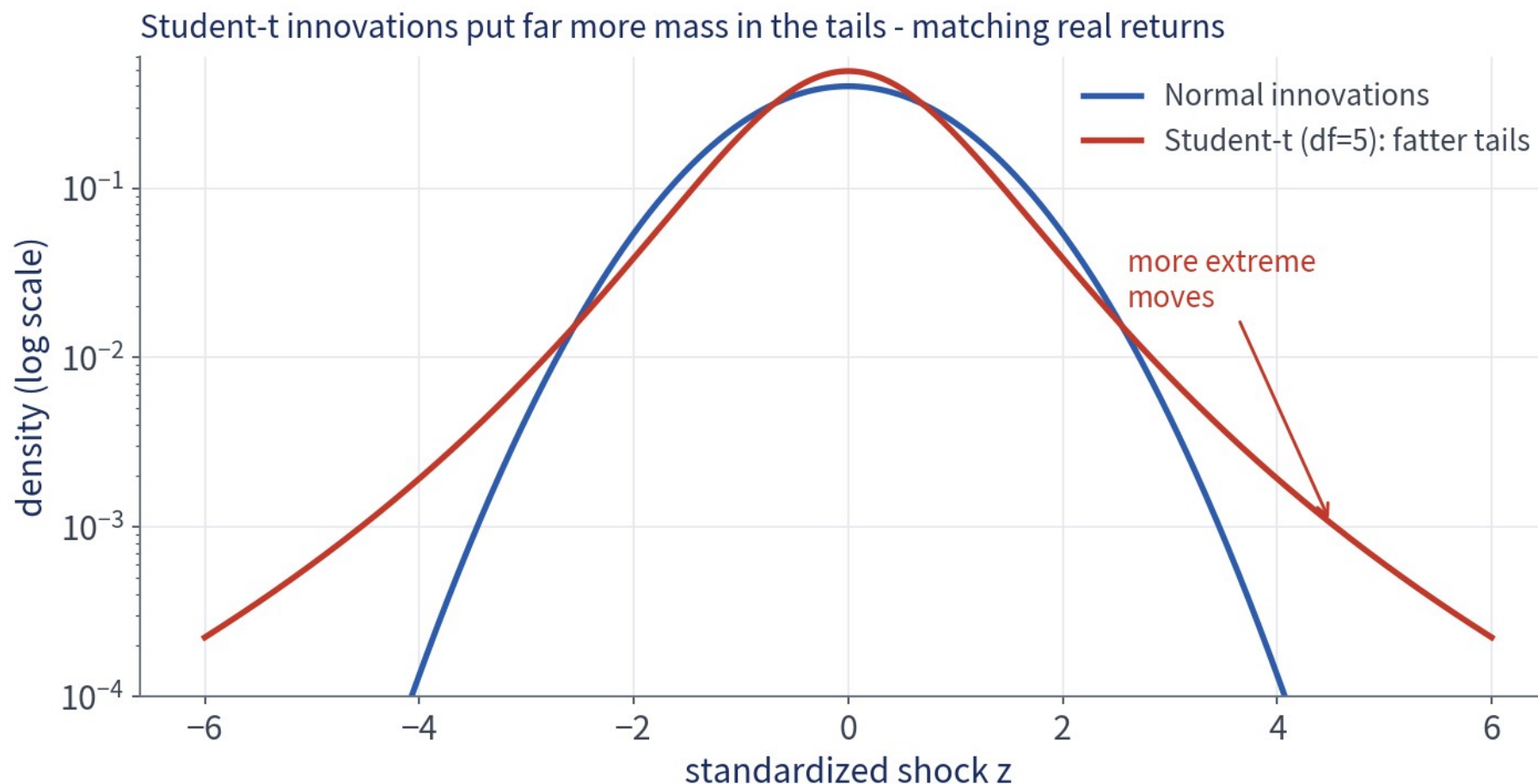
- GARCH already fattens the tails via changing variance.
- But real returns have even heavier tails than that.
- Extreme moves happen more often than normal shocks allow.

The fix

- Draw the standardized shocks z_t from a Student-t.
- The single parameter ν (degrees of freedom) sets tail weight.
- Lower ν means heavier tails — estimated from the data.

Normal vs Student-t Innovations

On a log scale the difference is stark: the Student-t places far more probability in the tails, matching the real frequency of extreme returns.



The Leverage Effect

Plain GARCH treats good and bad news the same

In equity markets, a price drop raises future volatility MORE than an equal-sized rise. Symmetric GARCH — which squares the shock — is blind to this asymmetry.

What we observe

- Bad news (a fall) spikes volatility sharply.
- Good news of the same size raises it far less.
- Squaring the shock erases the sign — and the effect.

Why it is called leverage

- A price fall raises a firm's debt-to-equity ratio.
- Higher leverage makes the equity riskier — more volatile.
- Whatever the cause, the asymmetry is robust and must be modeled.

GJR-GARCH: Asymmetry by an Extra Term

Let bad news add its own kick

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \underbrace{I_{t-1}}_{1 \text{ if } \varepsilon_{t-1} < 0} \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad \text{GJR-GARCH}$$

How it works

- An indicator switches on only when the last shock was negative.
- The extra γ term adds volatility just for bad news.
- Good and bad shocks now have different impacts.

Why it is popular

- It is a tiny, intuitive change to GARCH.
- $\gamma > 0$ confirms and measures the leverage effect.
- It is a strong default for equity volatility.

EGARCH: Asymmetry in Logs

Model the LOG of variance

$$\ln \sigma_t^2 = \omega + \alpha g(z_{t-1}) + \beta \ln \sigma_{t-1}^2 \quad \text{EGARCH (asymmetric, always positive)}$$

The idea

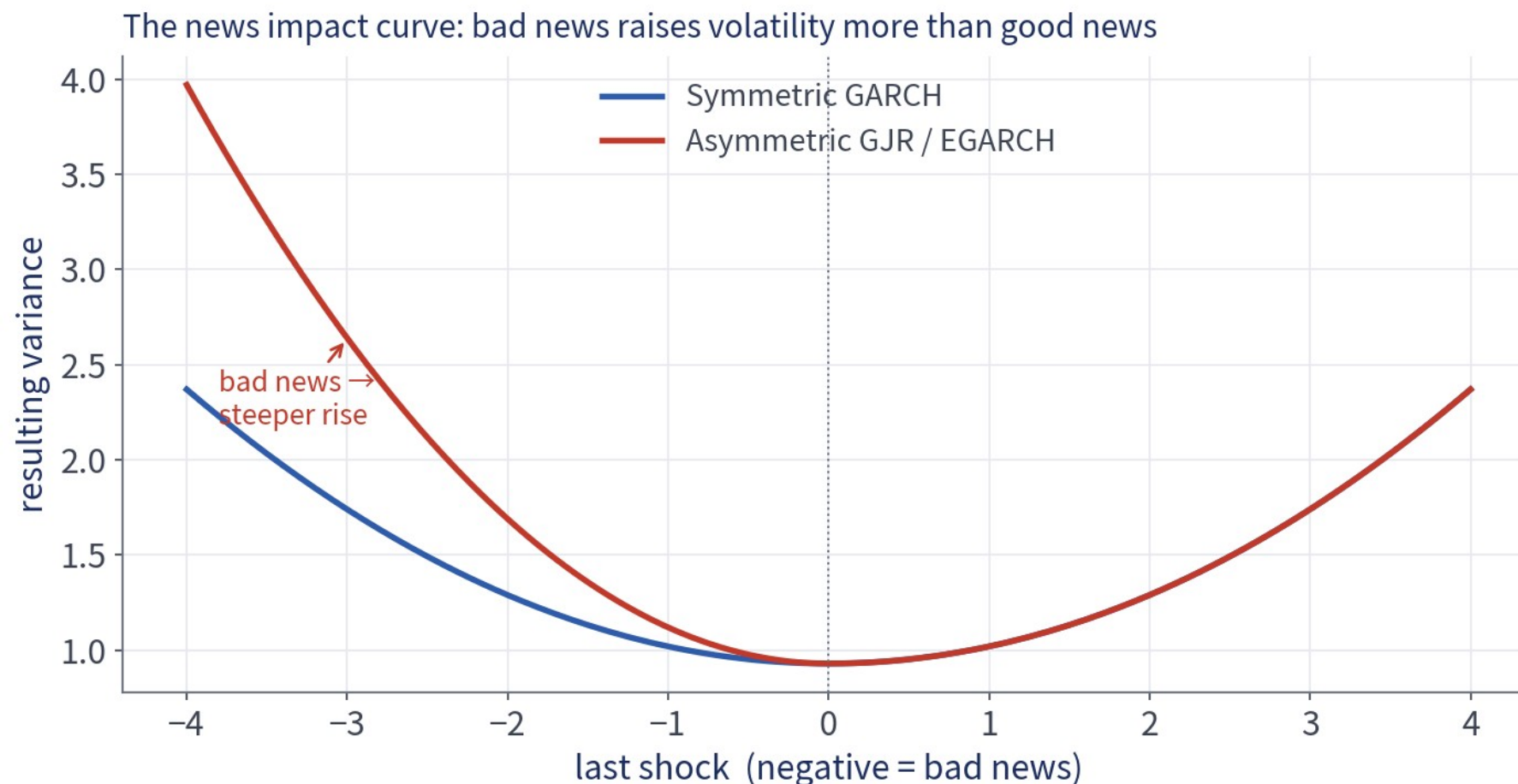
- Model the log of variance, so it is always positive.
- No non-negativity constraints on the parameters.
- Asymmetry enters through the shape of the news function.

GJR vs EGARCH

- Both capture the leverage effect well.
- EGARCH avoids positivity constraints; GJR is simpler to read.
- Try both and let the data (AIC/BIC) choose.

The News Impact Curve

How the last shock maps to today's variance. Symmetric GARCH is a clean parabola; GJR/EGARCH tilt it so bad news (left) raises volatility more than good news.



Choosing a Volatility Model

A short menu, in order of what to add

Model	What it adds	Use when
GARCH(1,1)	variance memory	the baseline — always start here
+ Student-t	heavier tails	extremes are more frequent than normal
GJR-GARCH	leverage / asymmetry	bad news raises volatility more (equities)
EGARCH	asymmetry, always positive	strong leverage; no positivity constraints

A common strong default for equities: a GJR- or E-GARCH(1,1) with Student-t innovations.

Estimating and Checking a GARCH

Fit by likelihood, validate on the residuals

Estimation

- Fit the mean and variance equations by maximum likelihood.
- Choose the innovation distribution (normal or Student-t).
- Confirm $\alpha + \beta < 1$ and all parameters are sensible.

Diagnostics

- Standardized residuals should be $\sim$ iid with no ARCH left.
- Check their squared ACF and re-run the ARCH-LM test.
- Compare candidates (GARCH/GJR/EGARCH) by AIC/BIC.

The Volatility Model Family

Start with GARCH(1,1); add Student-t innovations for fatter tails, and a leverage term (GJR/EGARCH) for asymmetry. Together they cover the stylized facts.

The Volatility Model Family

Start with GARCH, then add heavier tails and asymmetry as the data demands.



Fit the GARCH Family and Compare

Direct Claude Code across GARCH, GJR, and EGARCH

GOAL Fit a GARCH(1,1) and its extensions to real returns, forecast volatility, and let AIC/BIC pick the best.

[INPUT] Prompt · Claude Code

Write and run garch_lab.py that:

1. loads daily equity returns;
2. fits GARCH(1,1), GJR-GARCH, and EGARCH - each with normal AND Student-t innovations - using the 'arch' library;
3. reports alpha+beta (persistence), the half-life, and AIC/BIC for each;
4. plots the fitted conditional volatility and a 20-day volatility forecast for the winner, plus its news-impact curve.

Tell me which model wins and whether leverage and fat tails are significant.

How to work with it

- Expect persistence near 0.95–0.99 for equities.
- GJR/EGARCH usually beat plain GARCH on stocks.
- Student-t usually improves the fit noticeably.
- Keep it; Part 3 turns this into VaR and bands.

Deliverable: garch_lab.py + a model-comparison table and a volatility forecast.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1	A fitted GARCH(1,1) has $\alpha + \beta = 0.99$. What does that imply about risk?	Very persistent
Q2	Your standardized residuals still show fat tails. What extension helps?	Student-t innovations
Q3	Bad news raises volatility more than good news. Which models capture this?	GJR / EGARCH

All three come from persistence, the innovation distribution, and the leverage effect.

Three Pitfalls to Avoid

Where GARCH modeling goes wrong

Common mistakes

- "Higher-order GARCH always fits better."
- "Normal innovations are good enough."
- "Symmetric GARCH is fine for equities."

The reality

- GARCH(1,1) almost always suffices; higher orders rarely help.
- Real returns need Student-t tails — normal understates extremes.
- Equities show leverage; use GJR or EGARCH to capture it.

Three Things to Carry Forward

Part 2 in one screen

1

GARCH adds variance memory

One β term captures the long memory that ARCH needed many lags for. GARCH(1,1) — ω, α, β — is the three-parameter workhorse of risk modeling.

2

Persistence governs everything

$\alpha + \beta$ sets how slowly volatility reverts, its half-life, and its forecast term structure — near 1 means storms of risk fade slowly.

3

Two extensions complete the picture

Student-t innovations for fat tails, and GJR/EGARCH for the leverage effect — together they match every stylized fact of returns.

Next — Part 3: putting GARCH to work — Value-at-Risk, back-testing, and pairs-trading bands.